

Real-Branch Reversion at a Logarithmic Singularity

Finite log-power germs, irrational exponents,
rigorous remainders, and a Wolfram Language package

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Abstract

We construct the positive-real inverse at the singular endpoint zero of $f(x) = ax + \sum_j x^{1+\alpha_j} P_j(\log x)$, where $a > 0$, the finitely many α_j are positive real numbers, and the P_j are real polynomials. The construction answers both examples in the motivating Mathematica question: $x + x^2(1 + \log x)$ and $x + x^{\sqrt{2}}$. An auxiliary-parameter Lagrange–Bürmann formula is valid although ordinary Taylor reversion at zero is not. Euler differentiation reduces every coefficient to a finite polynomial calculation, and a locally finite additive semigroup determines exactly which irrational powers lie below a requested cutoff. We prove local real-branch existence, formal uniqueness, convergence for sufficiently small positive arguments, logarithm-aware cutoff remainders, and a computable conditional error majorant. The accompanying Wolfram Language package implements the coefficient formula, strict input validation, exact algebraic exponent comparison, independent sparse residual composition, and the error majorant. It includes examples and a native regression suite. Independent Python symbolic and numerical tests were executed; the native Wolfram suite was prepared but could not be executed because the available evaluator failed.

Principal correction to the motivating expectation. Writing $L = \log y$, the first example has

$$g(y) = y - y^2(1 + L) + y^3(2L^2 + 5L + 3) + \mathcal{O}(y^4(1 + |L|)^3).$$

Thus the two-term remainder is $\mathcal{O}(y^3 \log^2 y)$, not $\mathcal{O}(y^3)$.

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1 The problem and the branch that must be inverted

1.1 What the archived question asks

The question by Reshetnikov [1] begins with the real-valued function

$$f_{\text{real}}(x) = \begin{cases} x + \frac{x^2}{2} \log(x^2 e^2), & x \neq 0, \\ 0, & x = 0. \end{cases} \quad (1)$$

For positive x , this is simply

$$f_1(x) = x + x^2(1 + \log x). \quad (2)$$

An attempted combination of **InverseFunction** and **Series** produces expressions associated with another inverse branch, rather than the branch tending to zero. The update to the question asks also about

$$f_2(x) = x + x^{\sqrt{2}}, \quad x > 0. \quad (3)$$

In both cases the requested answer is a germ: a function $g(y)$ for small positive y , satisfying $f(g(y)) = y$ and $g(y)/y \rightarrow 1$. It is not an unrestricted global complex inverse.

The two obstructions are different but have a common solution. The logarithm prevents analyticity at zero. The irrational power prevents a usual power series or a Puiseux series with a fixed rational denominator. Neither obstruction prevents a convergent expansion in suitably chosen small parameters, nor a finite calculation to any prescribed exponent cutoff.

The answer by Michael E2 in the archived discussion introduces an auxiliary perturbation parameter, and the answer by Somos illustrates symbolic logarithmic reversion. We use this auxiliary-parameter idea as a starting point, but supply a branch theorem, an all-orders coefficient formula, an exact support algorithm, and remainders. The implementation here is newly written, not copied from those answers.

1.2 The positive branch is not the same as all real branches

For (2),

$$f_1'(x) = 1 + x(3 + 2 \log x).$$

The function $x(3 + 2 \log x)$ has its minimum at $x = e^{-5/2}$, with value $-2e^{-5/2}$. Consequently,

$$f_1'(x) \geq m_1 := 1 - 2e^{-5/2} > 0 \quad (x > 0). \quad (4)$$

Since $f_1(0+) = 0$ and $f_1(x) \rightarrow +\infty$, there is exactly one positive solution of $f_1(x) = y$ for every $y > 0$. The second example is also strictly increasing on the positive axis, since $f_2'(x) = 1 + \sqrt{2} x^{\sqrt{2}-1} > 0$.

This does *not* make (1) injective on the whole real line. For example, its negative tail tends to $+\infty$, so nonlocal negative solutions may coexist with the positive solution. Throughout the main construction, the branch condition is explicitly

$$y \rightarrow 0^+, \quad g(y) > 0, \quad g(y) \sim y/a. \quad (5)$$

The endpoint value $f(0) = 0$ fixes continuity; it does not supply missing higher derivatives or select a global inverse branch for a symbolic solver.

1.3 Why ordinary reversion is the wrong data structure

`InverseSeries` is documented for power-series data [4]. One should not infer that an ordinary Taylor series exists merely because a function has a first derivative at zero. In the first example, the singular logarithm occurs in higher derivatives. In the second, the correct exponents are $1 + n(\sqrt{2} - 1)$, not nonnegative integers.

We therefore represent a finite answer as a list of pairs $\{\beta, Q_\beta(L)\}$, interpreted as

$$\sum_{\beta} y^{\beta} Q_{\beta}(\log y),$$

together with separate remainder metadata. A request to retain all powers $\beta \leq B$ retains each associated logarithmic polynomial in full. In particular it does not silently turn $y^3 \log^2 y$ into $\mathcal{O}(y^3)$.

2 A finite log-power class and its ordered support

2.1 Normalization

We study

$$f(x) = ax + \sum_{j=1}^m x^{1+\alpha_j} P_j(\log x), \quad a > 0, \quad \alpha_j > 0, \quad P_j \in \mathbb{R}[L]. \quad (6)$$

Zero polynomials can be deleted, and equal exponents can be merged. Put

$$t = \frac{y}{a}, \quad Q_j = \frac{P_j}{a}, \quad F(x) = \frac{f(x)}{a} = x + h(x), \quad h(x) = \sum_j x^{1+\alpha_j} Q_j(\log x). \quad (7)$$

It suffices to solve $F(x) = t$. All coefficient formulas below refer to the normalized Q_j ; the final answer is evaluated at $t = y/a$, including $L = \log(y/a)$. Forgetting the factor a inside the logarithm is an error.

For $m > 0$, define

$$\delta = \min_j \alpha_j > 0, \quad d = \max_j \deg Q_j, \quad M(t) = 1 + |\log t|. \quad (8)$$

As $t \rightarrow 0^+$,

$$h(t) = \mathcal{O}(t^{1+\delta} M(t)^d), \quad h'(t) = \mathcal{O}(t^{\delta} M(t)^d) = o(1). \quad (9)$$

These estimates follow by differentiating each block, not by differentiating an unspecified big-O term. If the sum is empty, the inverse is exactly y/a .

Proposition 2.1 (Local positive-real inverse). *There exist $\varepsilon, \eta > 0$ such that F maps $(0, \varepsilon)$ strictly increasingly onto $(0, \eta)$, and its inverse G satisfies $G(t) = t + o(t)$. The inverse of f on the branch (5) is $g(y) = G(y/a)$.*

Proof. By (9), reduce ε so that $1/2 < F'(x) < 3/2$ on $(0, \varepsilon)$. Also $F(0+) = 0$. Strict monotonicity and continuity give the inverse. From $F(x)/x \rightarrow 1$, substituting $x = G(t) \rightarrow 0$ gives $G(t)/t \rightarrow 1$. $\square$

2.2 A locally finite, not necessarily rational, exponent grid

Let

$$S = \left\{ \alpha \cdot k = \sum_{j=1}^m \alpha_j k_j : k = (k_1, \dots, k_m) \in \mathbb{N}^m \right\}. \quad (10)$$

Here $\mathbb{N}$ includes zero. This is an additive semigroup, rather than the full additive group generated by the α_j .

Lemma 2.2 (Finite support below any cutoff). *For each $T \geq 0$, there are only finitely many multi-indices k with $\alpha \cdot k \leq T$. In particular $S \cap [0, T]$ is finite, and each $s \in S$ has only finitely many representations by such multi-indices.*

Proof. Every admissible multi-index satisfies $0 \leq k_j \leq \lfloor T/\alpha_j \rfloor$, and also $|k| \leq \lfloor T/\delta \rfloor$. These are finite bounds. $\square$

Irrationality does not imply accumulation at a finite positive cutoff here. For instance $\mathbb{N} + \sqrt{2}\mathbb{N}$ is locally finite, although the group $\mathbb{Z} + \sqrt{2}\mathbb{Z}$ is not. Confusing these two supports would incorrectly suggest that a finite-order answer is impossible.

2.3 Dominance and uniqueness of the blocks

For any fixed $\epsilon > 0$ and integer r , $t^\epsilon M(t)^r \rightarrow 0$. Thus every logarithmic polynomial at a larger power of t is smaller than every nonzero leading logarithmic monomial at a smaller power. Within a fixed power, a higher logarithm degree dominates a lower one. A nonzero polynomial has a uniquely determined leading degree. It follows, by subtracting two expansions and inspecting their first nonzero block, that the coefficients in a log-power expansion are unique.

The algebra used below consists of formal expressions $\sum_{s \in S} t^s C_s(L)$ with $C_s \in \mathbb{R}[L]$. Products are well-defined because each output weight has finitely many contributing pairs. The valuation is the least weight of a nonzero block; the zero expression has valuation $+\infty$. No numerical evaluation is used to decide equality or ordering of exponents in the package.

3 Reversion without analyticity at zero

3.1 The auxiliary parameter

For fixed positive t , introduce λ and solve

$$X + \lambda h(X) = t, \quad X(t, 0) = t. \quad (11)$$

The functions $X^{1+\alpha_j}$ and $\log X$ are holomorphic in a small complex disk centered at $t > 0$, using the logarithm inherited from the positive axis. Classical Lagrange–Bürmann inversion applies there [2, §1.10(vii)]; it is applied at $X = t$, not at the singular point $X = 0$. It gives

$$X(t, \lambda) = t + \sum_{n \geq 1} \frac{(-\lambda)^n}{n!} \frac{d^{n-1}}{dt^{n-1}} h(t)^n. \quad (12)$$

The derivative in this formula differentiates every occurrence of t , including $\log t$. Freezing the logarithm while reverting an ordinary polynomial would lose these derivative terms.

3.2 A convergence theorem and an explicit majorant

Fix $0 < r < 1$, and let $c_r = -\log(1 - r) > 0$. Write $Q_j(L) = \sum_{k=0}^{d_j} q_{jk} L^k$, and define

$$\mathcal{Q}_r(t) = \frac{1}{r} \sum_{j=1}^m t^{\alpha_j} (1+r)^{1+\alpha_j} \sum_{k=0}^{d_j} |q_{jk}| (|\log t| + c_r)^k. \quad (13)$$

This is a sufficient-condition majorant, not generally a sharp convergence boundary. Since every $\alpha_j > 0$, it tends to zero as $t \rightarrow 0^+$.

Theorem 3.1 (Analytic validity of auxiliary-parameter reversion). *For all sufficiently small positive t , the series (12) converges absolutely at $\lambda = 1$, and its sum is the branch $G(t)$ of Proposition 2.1. More generally, at any positive t with $\mathcal{Q}_r(t) < 1$, it sums to the unique root in $|X - t| < rt$, selected by continuation from $\lambda = 0$. This root is real and positive. For the locally defined branch, and for*

$$G_N(t) = t + \sum_{n=1}^N \frac{(-1)^n}{n!} D_t^{n-1} h(t)^n,$$

one has the explicit bound

$$|G(t) - G_N(t)| \leq \frac{tr}{N+1} \frac{\mathcal{Q}_r(t)^{N+1}}{1 - \mathcal{Q}_r(t)}. \quad (14)$$

The same bound holds at every point with $\mathcal{Q}_r(t) < 1$ for the just-described disk-selected root. Without a separate global monotonicity or continuation argument, that statement is not a claim about a globally continued origin branch outside its local domain. The expansion of each degree n into its finitely many multi-index contributions is absolutely summable over all multi-indices as well.

Proof. Set $X = t(1 + u)$, with $|u| \leq r$. The branch of $\log(1 + u)$ obtained from its Taylor series satisfies

$$|\log(1 + u)| \leq \sum_{k \geq 1} \frac{r^k}{k} = c_r.$$

Since all α_j are real, $|(1 + u)^{1+\alpha_j}| = |1 + u|^{1+\alpha_j} \leq (1 + r)^{1+\alpha_j}$. Consequently, on the circle $|X - t| = rt$,

$$\sum_j \sup |X^{1+\alpha_j} Q_j(\log X)| \leq rt \mathcal{Q}_r(t).$$

For $|\lambda| \mathcal{Q}_r(t) < 1$, Rouché's theorem compares $X - t + \lambda h(X)$ with $X - t$. There is exactly one zero inside this disk, counting multiplicities, hence it is simple. The implicit-function theorem continues that zero holomorphically in λ on the disk $|\lambda| < 1/\mathcal{Q}_r(t)$. The Taylor coefficients at zero are (12). For real $\lambda = 1$, complex conjugation and uniqueness make the zero real, and $X/t \in (1 - r, 1 + r)$ makes it positive. For sufficiently small t , this is precisely the local branch already identified.

Let M_j denote the individual suprema just used, and put $M = \sum_j M_j$. Cauchy's derivative estimate gives

$$\left| \frac{1}{n!} D_t^{n-1} h(t)^n \right| \leq \frac{(n-1)!}{n!(rt)^{n-1}} M^n \leq \frac{rt}{n} \mathcal{Q}_r(t)^n.$$

Summing for $n > N$, and replacing $1/n$ by $1/(N+1)$, proves (14). For absolute summability before merging multi-indices, apply Cauchy's estimate to $\prod_j h_j^{k_j}$ and sum over $|k| = n$. The

multinomial identity gives

$$\frac{(n-1)!}{(rt)^{n-1}} \sum_{|k|=n} \frac{\prod_j M_j^{k_j}}{\prod_j k_j!} = \frac{M^n}{n(rt)^{n-1}}.$$

The same summable geometric majorant results. Thus regrouping by exponent is legitimate, even when different multi-indices have the same weight. $\square$

Remark 3.2 (What “convergent” means here). The series is not an ordinary Taylor series in t . It is a convergent log-power series evaluated for sufficiently small positive t , and it is also an asymptotic expansion in the ordered log-power scale. Neither claim asserts analyticity at zero. Convergence of the auxiliary series does not imply that a default branch selected by `InverseFunction` is the branch being represented.

4 The finite coefficient formula

4.1 Euler differentiation of a block

For every real β and polynomial P ,

$$D_t(t^\beta P(L)) = t^{\beta-1}(\beta + \partial_L)P(L), \quad L = \log t. \quad (15)$$

Repeated differentiation gives

$$D_t^r(t^\beta P(L)) = t^{\beta-r} \prod_{j=0}^{r-1} (\beta - j + \partial_L)P(L). \quad (16)$$

All factors commute, since they are constant-coefficient polynomials in ∂_L . Their application never increases the logarithm degree.

For $k \in \mathbb{N}^m \setminus \{0\}$, write

$$n = |k|, \quad s = \alpha \cdot k, \quad k! = \prod_j k_j!, \quad Q^k(L) = \prod_j Q_j(L)^{k_j}.$$

Expanding h^n by the multinomial theorem and using (16) yields the central formula.

Theorem 4.1 (Multi-index inversion formula). *For the branch in (5),*

$$g(y) = t + \sum_{k \in \mathbb{N}^m \setminus \{0\}} t^{1+\alpha \cdot k} \frac{(-1)^{|k|}}{k!} (\partial_L + 2 + \alpha \cdot k)^{\overline{|k|-1}} \prod_j Q_j(L)^{k_j}, \quad t = \frac{y}{a}, \quad L = \log t. \quad (17)$$

An empty operator product equals the identity. This formula is formally well-defined coefficient-wise; it is absolutely convergent for sufficiently small positive y . Its power support is contained in $1 + S$, and the contribution from k has logarithm degree at most $\sum_j k_j \deg Q_j$.

Proof. The term of $h(t)^n$ indexed by k is $n! t^{n+s} Q^k(L)/k!$. Differentiating $n-1$ times lowers the power to $1+s$, and the operator factors are $\partial_L + n + s, \dots, \partial_L + 2 + s$. They are exactly the rising product in (17). The factor $n!$ cancels that in (12). Local finiteness follows from Lemma 2.2, and analytic validity from Theorem 3.1. $\square$

4.2 A completely scalar coefficient formula

Define numbers $r_{n,s,j}$ by

$$(z + 2 + s)^{\overline{n-1}} = \sum_{j=0}^{n-1} r_{n,s,j} z^j.$$

The block contributed by k is therefore

$$C_k(L) = \frac{(-1)^n}{k!} \sum_{j=0}^{\min(n-1, \deg Q^k)} r_{n,s,j} (Q^k)^{(j)}(L). \quad (18)$$

This provides a direct formula in finite products and polynomial coefficients, without symbolic differentiation of irrational powers. The numbers $r_{n,s,j}$ can equivalently be expressed through coefficients of rising factorials; that additional notation is not needed in the code.

For a single block $h = t^{1+\alpha} c L^q$, it specializes to

$$C_n(L) = \frac{(-c)^n}{n!} \sum_{j=0}^{\min(n-1, nq)} r_{n,n\alpha,j} (nq)^j L^{nq-j}.$$

For multiple generators, coincident weights must be *added*, not selected arbitrarily. These coincidences are the only “resonances” needed in this finite positive-exponent problem. There is no small-denominator inversion in (17).

5 Cutoffs, exact gaps, and rigorous remainders

5.1 What the cutoff requests

For an absolute power cutoff $B \geq 1$, set $T = B - 1$, and let g_B be (17) restricted to $\alpha \cdot k \leq T$, together with the leading term t . The least omitted semigroup point is

$$s_+(T) = \min\{s \in S : s > T\}. \quad (19)$$

It exists because S contains $\delta\mathbb{N}$, and it is computable from a finite enumeration: it lies in $(T, T + \delta]$.

Theorem 5.1 (Logarithm-aware exponent-cutoff remainder). *With $s_+ = s_+(T)$,*

$$g(y) - g_B(y) = \mathcal{O}\left(t^{1+s_+} M(t)^{d\lfloor s_+/\delta \rfloor}\right) \quad (t = y/a \rightarrow 0^+). \quad (20)$$

The exponent and logarithm degree in this bound are conservative if omitted coefficient blocks cancel. No nonzero-coefficient assumption is needed.

Proof. Choose an integer N with $(N+1)\delta > s_+$. The tail of the auxiliary-parameter series after degree N , by Theorem 3.1 and $\mathcal{Q}_r(t) = \mathcal{O}(t^\delta M^d)$, is

$$\mathcal{O}(t^{1+(N+1)\delta} M^{d(N+1)}) = o(t^{1+s_+}).$$

Every retained multi-index has $|k|\delta \leq T < s_+$, hence $|k| \leq N$. Among the finitely many degree- $\leq N$ contributions not retained, weights strictly larger than s_+ are $o(t^{1+s_+})$, even after multiplication by their fixed logarithmic polynomials. At weight exactly s_+ , $|k| \leq \lfloor s_+/\delta \rfloor$, so the logarithm degree is at most $d\lfloor s_+/\delta \rfloor$. Adding these finite contributions proves (20). $\square$

5.2 An explicit numerical-point bound for an exponent cutoff

The estimate above gives an asymptotic scale, not a constant or an endpoint threshold. A different, usually looser, bound supplies an explicit constant. Put $\alpha_{\max} = \max_j \alpha_j$ and

$$N_0 = \left\lfloor \frac{B-1}{\alpha_{\max}} \right\rfloor. \quad (21)$$

Every multi-index of total degree $n \leq N_0$ is retained, since $\alpha \cdot k \leq n\alpha_{\max} \leq B-1$. The omitted indices form a subset of the degrees $n > N_0$. Absolute multi-index summability in Theorem 3.1 therefore gives

$$\boxed{|g(y) - g_B(y)| \leq \frac{\text{tr } \mathcal{Q}_r(t)^{N_0+1}}{(N_0+1)(1 - \mathcal{Q}_r(t))}} \quad \text{provided } \mathcal{Q}_r(t) < 1. \quad (22)$$

This is the inequality implemented by `InverseErrorBound`. At an arbitrary evaluation point, the selected root is specified by $t(1-r) < X < t(1+r)$; near zero it is the requested germ. If the input is not globally monotone, using the bound for a different global branch requires additional justification. The two original examples are globally monotone on the positive axis, so this distinction causes no ambiguity there. It can be weak when the generators have very different sizes. It is nevertheless a valid bound for the requested exponent cutoff, not just for a homotopy truncation. A floating-point evaluation of its right-hand side is not an outward-rounded interval certificate; the package reports the condition explicitly and makes no such numerical certification claim.

5.3 Finite determination and remainder perturbations

The coefficient at weight s depends only on input blocks of weights $\leq s$. Hence a finite input jet determines a finite inverse jet. An analytic version requires a condition on the actual remainder, not only formal notation.

Proposition 5.2 (Stability under a controlled input remainder). *Suppose $F_0 = x + h(x)$ is as above, and $F(x) = F_0(x) + R(x)$, with*

$$R(x) = \mathcal{O}(x^{1+\rho}M(x)^q), \quad R'(x) = o(1), \quad \rho > 0.$$

Then both local positive inverses exist, and

$$F^{-1}(t) - F_0^{-1}(t) = \mathcal{O}(t^{1+\rho}M(t)^q).$$

Proof. Both derivatives tend to one, so $F' \geq 1/2$ after reducing the interval. Let $x_0 = F_0^{-1}(t) \sim t$. The residual of x_0 in the perturbed equation is $R(x_0)$. The mean-value theorem gives $|F^{-1}(t) - x_0| \leq 2|R(x_0)|$, and $x_0 \sim t$ transports the bound. $\square$

This proposition also explains why a finite log-power expansion cannot identify exponentially small additions such as $e^{-1/x}$. Such additions may leave every algebraic-logarithmic coefficient unchanged while changing the actual inverse by a flat term. They are not part of the finite exact input class accepted by this version of the package.

6 Formal composition and a second construction

6.1 The normalized correction equation

Write $G(t) = t(1 + u)$. Its equation is

$$u + \sum_j t^{\alpha_j} (1 + u)^{1+\alpha_j} Q_j(L + \log(1 + u)) = 0. \quad (23)$$

Since u has strictly positive valuation, both

$$(1 + u)^r = \sum_{n \geq 0} \binom{r}{n} u^n, \quad \log(1 + u) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} u^n \quad (24)$$

are well-defined formal series on the same semigroup. Only finitely many powers contribute to a fixed cutoff. Substitution into a polynomial Q_j is likewise finite after truncation.

Proposition 6.1 (Formal uniqueness and coefficient stabilization). *The map*

$$\mathcal{T}(u) = - \sum_j t^{\alpha_j} (1 + u)^{1+\alpha_j} Q_j(L + \log(1 + u))$$

has a unique positive-valuation fixed point. The iterates $u_0 = 0$, $u_{n+1} = \mathcal{T}(u_n)$, stabilize coefficientwise; after n iterations, $\text{val}(u - u_n) \geq (n + 1)\delta$.

Proof. For positive-valuation u, v , subtract their formal binomial and logarithm series. Each difference is divisible by $u - v$, with quotient of nonnegative valuation. The explicit factor t^{α_j} consequently implies

$$\text{val}(\mathcal{T}(u) - \mathcal{T}(v)) \geq \text{val}(u - v) + \delta.$$

Starting from zero gives $\text{val}(u_1 - u_0) \geq \delta$, and successive differences gain at least δ . By local finiteness, every coefficient stabilizes. The limit is a fixed point. Two different fixed points would have a finite first difference weight s , but the displayed inequality would force that same weight to be at least $s + \delta$, a contradiction. $\square$

Thus the Lagrange formula and fixed-point construction must give the same formal answer. This supplies an independent way to check coefficients and a conceptually simple alternative implementation.

6.2 Residual validation is not numerical root finding

For a finite correction u_B , the residual checker evaluates the left side of (23) by sparse multiplication and the finite jets (24). It does not call the Euler-operator formula that generated the answer. If the result vanishes through weight $B - 1$, then $f(g_B(y)) - y$ vanishes through absolute power B .

This statement is exact finite algebra. It does not automatically certify a floating-point root approximation. To obtain an actual error estimate from a numerical residual, one also needs a derivative lower bound and knowledge that the approximation and desired root lie in the chosen branch interval. If $f' \geq m > 0$ there, then

$$|g_B(y) - g(y)| \leq \frac{|f(g_B(y)) - y|}{m}. \quad (25)$$

A bracket $f(\ell) \leq y \leq f(r)$ on a strictly increasing branch gives $g(y) \in [\ell, r]$. For rigorous numerical use these inequalities must be checked exactly or with outward-rounded intervals.

For f_1 , the global bound (4) supplies $m = m_1$ for any positive approximation; for f_2 , one may use $m = 1$. These are stronger practical tools than a small raw residual alone.

7 The logarithmic example in full detail

7.1 All-order polynomial blocks

For $f_1(x) = x + x^2(1 + \log x)$, there is one generator $\alpha = 1$, and

$$g_1(y) = y + \sum_{n \geq 1} y^{n+1} C_n(L), \quad L = \log y, \quad (26)$$

where

$$C_n(L) = \frac{(-1)^n}{n!} (\partial_L + n + 2)^{\overline{n-1}} (1 + L)^n. \quad (27)$$

The first four polynomials are

$$\begin{aligned} C_1(L) &= -L - 1, \\ C_2(L) &= 2L^2 + 5L + 3, \\ C_3(L) &= -5L^3 - \frac{41}{2}L^2 - 27L - \frac{23}{2}, \\ C_4(L) &= 14L^4 + \frac{241}{3}L^3 + \frac{335}{2}L^2 + 151L + \frac{299}{6}. \end{aligned} \quad (28)$$

The next block, useful as an independent test fixture, is

$$C_5(L) = -42L^5 - \frac{3727}{12}L^4 - \frac{5365}{6}L^3 - 1256L^2 - \frac{5177}{6}L - \frac{2789}{12}. \quad (29)$$

For example, the $n = 2$ Lagrange term is

$$\frac{1}{2} D_y(y^4(1 + L)^2) = y^3(2(1 + L)^2 + (1 + L)),$$

which explains the $5L + 3$ terms that would be lost by incorrectly treating L as a constant during differentiation.

7.2 Catalan leading logarithms and the exact remainder scale

Only the zero-derivative part of the operator contributes to degree n in C_n , so

$$[L^n]C_n(L) = \frac{(-1)^n}{n!} (n + 2)^{\overline{n-1}} = (-1)^n \frac{1}{n+1} \binom{2n}{n}. \quad (30)$$

These are signed Catalan numbers, and they are nonzero. It follows that

$$g_1(y) - \left(y + \sum_{n=1}^N y^{n+1} C_n(\log y) \right) = \mathcal{O}(y^{N+2} M(y)^{N+1}), \quad (31)$$

with that logarithmic degree actually present in the first omitted block. In particular,

$$\frac{g_1(y) - y + y^2(1 + \log y)}{y^3} = 2 \log^2 y + 5 \log y + 3 + o(1),$$

where the $o(1)$ follows from the next remainder divided by y^3 . The ratio is unbounded, proving that $\mathcal{O}(y^3)$ is not the correct remainder for the two displayed leading terms.

7.3 An analytic two-parameter interpretation

The correction equation becomes

$$u + y(1 + u)^2(1 + L + \log(1 + u)) = 0.$$

Introduce $v = y(1 + L)$ and $w = y$. This is

$$u + (1 + u)^2(v + w \log(1 + u)) = 0. \quad (32)$$

The analytic implicit-function theorem at $(u, v, w) = (0, 0, 0)$ gives a convergent two-variable power series $u = U(v, w)$. Since both $y(1 + \log y)$ and y tend to zero, evaluating this analytic function along that curve produces (26). The logarithmic singularity of the curve is compatible with analyticity of the lifted function.

A sharper one-example majorant than (13) is obtained by using $|1 + L|$ directly. For $r = 1/2$, set

$$q(y) = \frac{9}{2}y(|1 + \log y| + \log 2). \quad (33)$$

If $q(y) < 1$, Cauchy's estimate gives

$$|g_1(y) - g_{1,N}(y)| \leq \frac{y}{2(N+1)} \frac{q(y)^{N+1}}{1 - q(y)}. \quad (34)$$

This is a sufficient, explicitly checkable domain, not a claim that the series ceases to converge whenever the inequality fails. The generic package majorant is slightly looser because it sums absolute polynomial coefficients.

7.4 The negative-side germ of the original real function

For $x = -u$, $u > 0$, and $y = -v$, the equation $f_{\text{real}}(x) = y$ is

$$v = u - u^2(1 + \log u).$$

It is another member of our class, with polynomial $-(1 + L)$. Substituting back shows that the local two-sided inverse can be written with the same integer powers and with $\log |y|$:

$$g_{\text{real}}(y) = y + \sum_{n \geq 1} y^{n+1} C_n(\log |y|), \quad y \neq 0$$

for sufficiently small $|y|$. To compute the negative side using the positive-only package, invert $u - u^2(1 + \log u)$ for $v > 0$ and then set $y = -v$, $g(y) = -u(v)$. The package never globally rewrites $\log(x^2)$ as $2 \log x$ without the positive-real assumption.

8 The irrational-power example and its convergence radius

8.1 A one-generator generalized power series

Consider the entire family $f_p(x) = x + x^p$, with $p > 1$, and put $\alpha = p - 1$. The inverse is

$$g_p(y) = \sum_{n \geq 0} c_n(p) y^{1+n\alpha}, \quad c_0(p) = 1, \quad (35)$$

where the main theorem gives, for $n \geq 1$,

$$\begin{aligned} c_n(p) &= \frac{(-1)^n}{n!} \prod_{j=0}^{n-2} (np - j) \\ &= \frac{(-1)^n \Gamma(np + 1)}{\Gamma(n + 1) \Gamma(n(p - 1) + 2)} = \frac{(-1)^n}{1 + n(p - 1)} \binom{np}{n}. \end{aligned} \quad (36)$$

The empty product for $n = 1$ is one. The finite product, rather than gamma quotients, is used for symbolic coefficient generation in the package.

The first terms for general p are

$$\begin{aligned} g_p(y) &= y - y^p + p y^{2p-1} - \frac{p(3p-1)}{2} y^{3p-2} \\ &\quad + \frac{p(4p-1)(4p-2)}{6} y^{4p-3} + \mathcal{O}(y^{5p-4}). \end{aligned} \quad (37)$$

For the requested value $p = \sqrt{2}$, this becomes

$$\begin{aligned} g_2(y) &= y - y^{\sqrt{2}} + \sqrt{2} y^{2\sqrt{2}-1} - \left(3 - \frac{\sqrt{2}}{2}\right) y^{3\sqrt{2}-2} \\ &\quad + \left(\frac{17\sqrt{2}}{3} - 4\right) y^{4\sqrt{2}-3} + \mathcal{O}(y^{5\sqrt{2}-4}). \end{aligned} \quad (38)$$

There are finitely many terms below every exponent cutoff. To retain terms through the $n = N$ contribution, pass the exact cutoff $B = 1 + N(\sqrt{2} - 1)$, not a decimal approximation to it.

8.2 Why this is not a Puiseux series

If p is rational, a suitable root of y makes all exponents integral. If p is irrational, no fixed rational denominator does so. Nevertheless, $z = y^{p-1}$ makes the correction an ordinary analytic series:

$$g_p(y) = yW(z), \quad W(z) + zW(z)^p = 1, \quad W(0) = 1.$$

The implicit derivative with respect to W is one at $(W, z) = (1, 0)$, which immediately proves a nonzero radius of convergence in z .

8.3 Exact radius from the coefficient formula

Stirling's formula applied to (36) gives

$$|c_n(p)| \sim \frac{\sqrt{p}}{\sqrt{2\pi}(p-1)^{3/2}} n^{-3/2} \rho^{-n}, \quad \rho = \frac{(p-1)^{p-1}}{p^p}. \quad (39)$$

Indeed, the exponential factors cancel between the three gamma functions; the remaining exponential base is $p^p/(p-1)^{p-1}$, and the power of n is $-3/2$. Thus the radius of $W(z)$ is exactly ρ . On the positive axis, (35) certainly converges for

$$0 < y < \rho^{1/(p-1)}. \quad (40)$$

The coefficient estimate also gives absolute convergence at $|z| = \rho$, although analyticity need not extend through the negative boundary point. For $p = \sqrt{2}$, numerical evaluation gives

$$\rho \approx 0.4251956066518408138025026, \quad \rho^{1/(p-1)} \approx 0.1268627051233096813979122.$$

The series boundary is not a boundary of existence of the positive inverse: that inverse exists for every $y > 0$. It is a boundary for this expansion about the chosen auxiliary coordinate $z = 0$.

9 Mixed generators and coincident exponents

9.1 The first mixed term

For

$$F(x) = x + x^{1+\alpha}P(\log x) + x^{1+\beta}Q(\log x),$$

the contributions with multi-indices $(1, 0)$, $(0, 1)$, $(1, 1)$ are

$$-t^{1+\alpha}P(L), \quad -t^{1+\beta}Q(L), \quad t^{1+\alpha+\beta}(\partial_L + 2 + \alpha + \beta)(P(L)Q(L)).$$

The two pure quadratic contributions are

$$\frac{1}{2}t^{1+2\alpha}(\partial_L + 2 + 2\alpha)P(L)^2, \quad \frac{1}{2}t^{1+2\beta}(\partial_L + 2 + 2\beta)Q(L)^2.$$

These terms must be sorted by their actual exponents, not by total multi-index degree. For example, a contribution of degree three in a small generator can precede a contribution of degree one in a large generator.

9.2 A resonant worked example

Take $\alpha = 1/2$, $\beta = 1$, $P(L) = 1 + L$, and $Q(L) = 2 - L$. The exponent $1 + \beta = 2$ coincides with $1 + 2\alpha = 2$; both contributions are required. The result is

$$\begin{aligned} G(t) = t - (1 + L)t^{3/2} + \left(\frac{3}{2}L^2 + 5L + \frac{1}{2}\right)t^2 \\ + \left(-\frac{21}{8}L^3 - \frac{123}{8}L^2 - \frac{123}{8}L + \frac{3}{8}\right)t^{5/2} + \mathcal{O}(t^3M(t)^4). \end{aligned} \quad (41)$$

The remainder shown uses the conservative general theorem. The package merges equal input generators before enumeration and equal output exponents afterward. It also deletes blocks whose combined polynomial is zero.

Even an ordinary polynomial exposes the cancellation issue. For $F = x + x^2 + 2x^3$, the inverse has $t - t^2 + 0t^3 + \dots$, whereas $F = x + x^2 + 3x^3$ has $t - t^2 - t^3 + \dots$. A data structure that overwrites a coincident exponent rather than adding its coefficients fails these tests.

10 The Wolfram Language implementation

10.1 Supported contract and public entry points

The mathematical theory permits arbitrary fixed positive real generators. The software deliberately restricts generators and the cutoff to exact real algebraic numbers. This makes ordering, equality, and bounded lattice enumeration effectively decidable with **RootReduce** and exact symbolic comparisons [5, 6]. Polynomial coefficients and the positive linear coefficient may be exact, provably real numeric constants, such as rationals, radicals, or π ; unresolved symbolic parameters and machine-precision constants are not accepted.

The main public interfaces are:

```
RealInverseExpansion[f, {x,y}, B]
LogPowerInverseExpansion[{{alpha1,P1}, ...}, {y,L}, B]
InverseResidual[data, B]
InverseErrorBound[data, y0]
PurePowerInverseCoefficient[p, n]
LogQuadraticInversePolynomial[n, L]
```

The structured interface accepts the blocks of the original, unnormalized $f: x^{1+\alpha_j} P_j(\log x)$. Its option "LinearCoefficient" $\rightarrow a$ performs the division by a internally. The supplied logarithm symbol L must differ from the output symbol. The expression front end normalizes under $x > 0$ and then parses finite monomials. It never uses an unrestricted **PowerExpand**.

Clear the symbols used as variables before calling the package. As with ordinary Wolfram Language algebra, pre-existing values of x , y , or the logarithm symbol can alter an expression before it reaches the function. The documented interfaces accept expressions, not held unevaluated programs.

10.2 Loading and the two original examples

From the extracted archive root:

```
Get["Kernel/RealLogPowerInverse.wl"];
Clear[x,y,L];

r1 = RealInverseExpansion[x+x^2 (1+Log[x]), {x,y}, 5];
r1["Expression"]
r1["Remainder"]
InverseResidual[r1] ["Vanishes"]

r2 = RealInverseExpansion[x+x^Sqrt[2],
                          {x,y}, 1+6 (Sqrt[2]-1)];
r2["Expression"]
InverseResidual[r2] ["Vanishes"]
```

The residual outputs are expected to be **True**; these exact calls are included in the native regression suite. They are not represented here as observed native-kernel output, because that suite was not executed in this environment.

The structured equivalent of the first call is

```
r1 = LogPowerInverseExpansion[{{1,1+L}}, {y,L}, 5];
```

For the original **Piecewise** definition, one can first simplify the positive restriction:

```
f = Piecewise[{{x+x^2 Log[x^2 E^2]/2, x!=0}}, 0];
fPositive = FullSimplify[f, Assumptions -> x>0];
RealInverseExpansion[fPositive, {x,y}, 3]
```

The explicit structured interface remains available when the expression front end cannot normalize a semantically equivalent input form.

10.3 Returned data and remainder semantics

Each expansion is an **Association**. The most important fields are

Field	Meaning
"Expression"	The finite expression in y/a and $\log(y/a)$.
"Terms"	Pairs $\{\beta, C(L)\}$, with absolute powers β .
"CorrectionBlocks"	Pairs $\{s, C(L)\}$ representing $G(t)/t - 1$.
"InputBlocks"	Normalized pairs $\{\alpha_j, P_j(L)/a\}$.
"Remainder"	Power, logarithm degree, scale, and the asymptotic interpretation.
"MultiIndexCount"	Number enumerated, including the zero index and the finite next-gap lookahead.

The remainder scale is an ordinary expression accompanied by a statement of its big-O meaning. It is not a `SeriesData` object with an invented integer order. Its implicit constant and threshold are not supplied. For a computable conditional bound at an exact point, use, for example,

```
b = InverseErrorBound[r1, 1/1000];
b["Condition"]
b["Bound"]
```

The bound is a `ConditionalExpression` with condition $Q_r < 1$, using "Radius" $\rightarrow 1/2$ by default. Failure of this sufficient condition is not evidence that the expansion is invalid.

10.4 Core algorithm and termination

The coefficient engine performs the following finite calculation. It first validates and merges the normalized input blocks. It then enumerates $k \in \mathbb{N}^m$ with $\alpha \cdot k \leq B - 1 + \delta$ recursively, using the coordinate bound

$$k_j \leq \left\lfloor \frac{B - 1 + \delta - \sum_{i < j} \alpha_i k_i}{\alpha_j} \right\rfloor.$$

The extra δ finds the next semigroup point for the remainder. For every retained nonzero index it constructs Q^k , applies the Euler factors, divides by $k!$, and merges the result by exact exponent equality. The essential polynomial loop is

```
q = Times @@ MapThread[Power, {polynomials, k}];
Do[q = Expand[(2+s+j) q + D[q,L]], {j,0,n-2}];
coefficient = (-1)^n q / (Times @@ (Factorial /@ k));
```

A constant term in a polynomial is handled by the same code; for $n = 1$ the loop is empty. Factorial normalization is multi-index factorial $\prod k_j!$, not $|k|!$, because the multinomial coefficient has already been accounted for.

The number of lattice points is bounded by $\prod_j (1 + \lfloor (B - 1 + \delta)/\alpha_j \rfloor)$. This bound can be large; it is not a polynomial-time promise in the number of generators. The option "MaxMultiIndices" $\rightarrow 100000$ stops excessive enumeration with a `Failure` object. The independent residual checker has its own "MaxProducts" $\rightarrow 1000000$ guard. Exact algebraic comparison and polynomial simplification can still be costly; these guards count algebraic work units rather than elapsed time or total memory.

10.5 Independent sparse composition

The residual engine stores a sparse series as a sorted list of $\{s, P(L)\}$ pairs. Addition merges coefficients, multiplication adds weights and discards terms past the cutoff, binomial and logarithm series stop when their next power has no remaining support, and polynomial substitution uses Horner's rule. All correction weights must be positive; this is what ensures the loops terminate.

The result includes **"NormalizedResidual"**, representing $(f(g_B(y)) - y)/y$, and **"Residual"**, representing the unnormalized residual after truncation. The checker verifies the polynomial data in **"CorrectionBlocks"**; hand-editing only **"Expression"** does not change that data and is outside the supported workflow. Treat returned associations as immutable records, or rebuild them consistently.

10.6 Deliberate limitations

The implementation is not a universal transseries system. It rejects nonpositive excess exponents, iterated logarithms, negative logarithm powers, oscillatory coefficients, approximate or nonalgebraic exponent data, and unspecified input remainders. It does not determine all global inverse branches, prove arbitrary user assumptions, or formally verify itself in a proof assistant. The two requested examples and their finite mixed-log-power generalizations lie fully inside its supported class.

The two coefficient helper functions return formal symbolic polynomials and are not branch validators. For example, `PurePowerInverseCoefficient[p,n]` is useful symbolically even when `p` is a parameter; the inverse-function theorem for that family still requires $p > 1$. The main expansion interfaces enforce their stricter numeric-domain contract.

11 Verification, numerical evidence, and reproducibility

11.1 What was actually executed

The independent script `validation/reference_validation.py` was run with SymPy and mpmath. It completed 19 grouped symbolic checks and 27 high-precision numerical cases. The symbolic checks include direct residual composition, fixed-point cross-checks, mixed irrational and resonant support, scaled coefficients, noninteger cutoffs, Catalan leading coefficients, ordinary quadratic inversion, and detection of a deliberately corrupted coefficient. Its polynomial calculations are exact, not fitted to numerical samples.

The numerical comparisons were performed at 110 decimal digits. The reported errors compare truncated expansions with high-precision positive roots. They are numerical evidence, not interval-arithmetic certificates. The logarithmic cases additionally check the numerical residual bound using m_1 from (4).

A genuine Wolfram evaluator call failed with an unavailable server/dependency error, and no local Wolfram kernel was present. Consequently the delivered Wolfram regression suite was *not* executed. This is an explicit release limitation: independent mathematical validation does not establish native Wolfram execution, parser behavior, or compatibility on every claimed target version. The full package source and tests are supplied so those checks are reproducible in the user's installation.

11.2 Representative numerical results

Here N is the number of nonleading contributions retained in (26) or (35), not an absolute exponent cutoff. The ratio compares the absolute error with the magnitude of the first omitted term.

Example	y	N	Absolute error	Error/next term
Logarithmic	10^{-2}	1	$2.42021 \cdot 10^{-5}$	1.08096
Logarithmic	10^{-2}	6	$1.07915 \cdot 10^{-10}$	1.10472
Logarithmic	10^{-3}	3	$1.25941 \cdot 10^{-11}$	1.01590
Logarithmic	10^{-6}	1	$3.15669 \cdot 10^{-16}$	1.00003
Logarithmic	10^{-6}	6	$1.79594 \cdot 10^{-38}$	1.00004
Irrational power	10^{-2}	1	$2.51975 \cdot 10^{-4}$	0.80853
Irrational power	10^{-2}	6	$3.37743 \cdot 10^{-7}$	0.77205
Irrational power	10^{-3}	3	$3.88767 \cdot 10^{-8}$	0.90508
Irrational power	10^{-6}	1	$1.50541 \cdot 10^{-11}$	0.99473
Irrational power	10^{-6}	6	$1.09682 \cdot 10^{-22}$	0.99351

The ratios tending to one as y decreases agree with the nonzero leading omitted-block analysis. The irrational-power example approaches its small parameter regime more slowly because $y^{\sqrt{2}-1}$ decreases more slowly than y .

11.3 Running the native tests and rebuilding the article

The native suite contains 34 `VerificationTest` cases. It exercises expression and structured interfaces, both examples, exact collisions and cancellations, scaling, explicit error bounds, invalid input, resource limits, and detection of corrupted correction data. Wolfram's `TestReport` interface is documented in [7]. From the archive root run

```
TestReport["Tests/RealLogPowerInverse.wlt"]
```

or from a terminal with a working kernel:

```
wolframscript -file Tests/RunTests.wls
python validation/reference_validation.py
cd article
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command is intentionally a user-run native test, not a report of successful execution here. The Python output is recorded in `validation/results.json` and `validation/run.log`. The bibliography is embedded in this source; no external bibliography file or network access is required to build the PDF.

12 Relation to the supplied research materials

The supplied ProveIt materials organize polynomial-logarithmic scales, well-based support, differential block operators, reversion, and remainder transport [3]. Their orientation and status files explicitly distinguish classical coefficient Lagrange–Bürmann inversion from near-identity operator formulations, and distinguish formal identities from analytic branch and convergence

claims. Those distinctions are important here: the passage from (12) to (17) is proved, not inferred from the shared name “Lagrange inversion.”

The connection is concrete. The block derivative is $t^{\beta-1}(\beta + \partial_L)$; the positive semigroup supplies finite factorization and cutoff termination; composition closes on the same support; and a real derivative lower bound transports residuals to actual inverse errors. The present setting is intentionally smaller than a full transseries field. Its positive, finite generators permit elementary proofs and a compact implementation, without importing the full machinery of well-based transseries.

Scope of source inspection

The repository reference is pinned to the following commit:

24ce8bd743eaab64a91ce90725ea00f498d319d2.

The group and consolidated-volume README files and targeted search excerpts were accessible. The oversized consolidated $\text{\TeX}$ volume could not be retrieved in full through the available connector. Accordingly, this article does not claim a line-by-line review of that volume or verification of its formalization inventory. All essential mathematical arguments used by the package are stated and proved here.

13 Extensions and boundaries of the method

13.1 Other endpoints and nonunit leading powers

A problem at infinity may sometimes be moved to zero by reciprocal coordinates, but the transformed equation must still be checked for positive excess exponents and a nonzero linear leading term. The direction of the asymptotic ordering changes under such substitutions; it should not be handled by merely reversing a sorting routine.

For a leading term ax^β , $a, \beta > 0$, one can first set $t = (y/a)^{1/\beta}$. Taking the positive β -th root of the normalized forward equation produces a near-identity equation. The root usually creates an infinite binomial expansion rather than a finite exact log-power sum. Finite determination and a controlled remainder, as in Proposition 5.2, then justify finite-jet inversion. That preprocessing is a mathematical extension, not an automatic feature of this release.

13.2 Faster engines without changing the specification

The multi-index coefficient formula favors transparency. At high cutoffs, one could replace repeated polynomial products by cached powers, use a priority queue for support enumeration, or use truncated Newton iteration on (23). Its derivative with respect to u is a unit 1+ positive-valuation terms, so formal inversion is possible and Newton steps increase the accuracy rapidly. These optimizations should preserve the same exact cutoff and remainder semantics, and should still be checked by a separate composition engine.

For many mutually dependent generators, a semigroup normal form or lattice relation basis could avoid repeated representations. It must not discard multiplicity: different representations contribute distinct terms in (17). Any optimized enumeration must reproduce their sum.

13.3 What fails outside the hypotheses

If $\alpha_j = 0$, a logarithmic correction need not be small relative to the linear term, and the positive-valuation iteration can fail. If infinitely many positive generators accumulate at zero, local finite enumeration fails. Negative logarithm powers require rational-function rather than polynomial coefficients and different zero-weight bookkeeping. Oscillatory terms can require an enlarged coefficient algebra and separate branch estimates. These cases need genuinely additional hypotheses; accepting them silently would undermine both termination and the analytic error claims.

14 Conclusion

The requested inverses do not require a Taylor series at the singular endpoint. They require a branch condition and the right asymptotic algebra. For finite positive log-power perturbations of a linear germ, the answer is fully constructive: normalize the linear coefficient, enumerate a finite positive exponent simplex, apply the Euler-polynomial Lagrange formula, merge coincident weights, and report a logarithm-aware remainder. Analytic continuation in an auxiliary parameter validates the formal answer and provides an explicit conditional error majorant. The two examples are special cases of the same theorem, with Catalan leading logarithms in one case and generalized binomial coefficients on an irrational grid in the other.

A Archive contents and provenance

Path	Purpose
article/article.tex, article.pdf	This source and compiled article.
Kernel/RealLogPowerInverse.wl	Main Wolfram Language package.
Kernel/init.m, PacletInfo.wl	Loading and package metadata.
Examples/Examples.wl	Executable usage examples.
Tests/RealLogPowerInverse.wlt	Native regression tests, not executed here.
Tests/RunTests.wls	Command-line native test runner.
validation/reference_validation.py	Executed independent symbolic/numerical checks.
validation/results.json, run.log	Recorded independent results.
README.md, VALIDATION.md	Usage and explicit testing status.
LICENSE, SHA256SUMS	License for newly written code; checksums.

The user's source archive was consulted but is not repackaged here. Public sources are cited for attribution and context. The mathematical proofs, package, and independent test implementation in this archive were prepared for this task. The code license does not purport to relicense any cited third-party material.

References

- [1] V. Reshetnikov, *How to get an asymptotic of the real-valued branch of the inverse function?*, Mathematica Stack Exchange, question 236367, December 13, 2020; with answers by Michael E2 and Somos. <https://mathematica.stackexchange.com/questions/236367/>. The user-supplied archive and the public page were consulted.

- [2] NIST Digital Library of Mathematical Functions, §1.10(vii), *Inverse Functions*, especially the Lagrange inversion formulas. <https://dlmf.nist.gov/1.10#vii>.
- [3] V. Reshetnikov and contributors, *ProveIt*, polynomial-logarithmic series and transseries materials, commit 24ce8bd743eaab64a91ce90725ea00f498d319d2. <https://github.com/VladimirReshetnikov/ProveIt/tree/24ce8bd743eaab64a91ce90725ea00f498d319d2/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>. See the source-inspection limitation in Section 12.
- [4] Wolfram Research, *InverseSeries*, Wolfram Language documentation. <https://reference.wolfram.com/language/ref/InverseSeries.html>.
- [5] Wolfram Research, *RootReduce*, Wolfram Language documentation. <https://reference.wolfram.com/language/ref/RootReduce.html>.
- [6] Wolfram Research, *Algebraics*, Wolfram Language documentation. <https://reference.wolfram.com/language/ref/Algebraics.html>.
- [7] Wolfram Research, *TestReport*, Wolfram Language documentation. <https://reference.wolfram.com/language/ref/TestReport.html>.

Online sources were consulted on September 7, 2026.